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Research Paper

Development of a peer support and precision matching mobile platform for health management for people living with HIV

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ABSTRACT

Objectives: This study aimed to describe the development process of the peer precision-matching and health management mobile platform "Aspark" for people living with human immunodeficiency virus (HIV) (PLWH) and to evaluate its usability.

Methods: The study assembled a multidisciplinary research and development team comprising researchers, software engineers, art designers, clinical nursing specialists, HIV peer volunteers, and psychological counselors. The team employed an agile development model to iteratively build the platform. Using the social determinants of health framework and the interpersonal circumplex, we identified 14 matching variables for the mobile platform. A video production team was formed to create peer training videos on health management for PLWH. We integrated relevant tools to provide PLWH with various types of support, including informational, instrumental, emotional, affiliational, and appraisal assistance. From March to May 2024, the research team used convenience sampling to select 130 platform users. It administered an electronic questionnaire based on the System Usability Scale to evaluate the platform's overall usability.

Results: The platform incorporates multiple functional modules, including "Precision Matching," "Drug Interaction Query," "Medication Reminder," "Health Management," "Mood Diary," "Community Discussion," and "Activity Center," achieving comprehensive coverage across informational, emotional, instrumental, social, and appraisal support. The precision matching module facilitates intelligent pairing between patients and peer volunteers; the health management module allows for the entry of test results and trend analysis; medication reminders and check-in features enhance medication adherence; the mood diary supports emotional recording and expressive writing; while the community module offers functions such as topic discussions, appointment scheduling, and organization of offline activities. Among the 130 valid questionnaires, more than 83 % of users scored above 68 on the System Usability Scale, indicating that the platform demonstrates favourable usability.

Conclusions: The "Aspark" platform offers a feasible and promising digital solution for precision matching in HIV peer support and health management, enabling healthcare providers to deliver personalized and continuous health support to patients through its functions. Well-designed clinical trials should be conducted in the future to evaluate the effectiveness of this tool.

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What is known?

- Peer support plays a vital role in the health management of people living with HIV (PLWH) as an essential complement to medical services, and the effectiveness and sustainability of peer support needs to be improved.
- The effectiveness and sustainability of HIV peer support can be improved by accurately matching PLWH with peer volunteers.
- There is currently no mobile platform that provides peer support for PLWH and is designed to accurately match PLWH with peer volunteers, which can provide practical health management tools for PLWH.

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What is new?

- We developed a platform ("Aspark") that includes a matching algorithm between PLWH and peer volunteers to stimulate the potential significance of peer support for HIV health management.
- The evaluation results of the platform's usability show that users perceive the platform to have a good level of effectiveness and efficiency, indicating its potential as a practical tool for PLWH.

1. Introduction

Since the development of antiretroviral therapy (ART), people living with human immunodeficiency virus (HIV) (PLWH) can achieve a relatively satisfying life expectancy [1]. PLWH often endure numerous symptoms and have unmet care needs [2], typically experiencing a range of both physical and psychological manifestations [3]. Common symptoms experienced by PLWH encompass fatigue, depression, pain, rash, and insomnia, with an average of 8–17 symptoms being reported [4,5]. However, objectively, medical resources are inadequate to address the intricate care needs of PLWH, and HIV-related discrimination further hinders their access to medical and psycho social support [6]. Given the aforementioned issues, novel breakthroughs must be pursued to attain the "95-95-95" targets [7].

Given that peer support has enormous potential human resources, irreplaceable advantages in dealing with barriers to HIV-related discrimination, and potential comprehensive benefits for PLWH, peer support is a significant breaking point [8]. As the concept of peer support continues to gain traction, its significance and worth have garnered increasing recognition. The Joint United Nations Programme on HIV/AIDS (UNAIDS) has also emphasized the importance of fostering the involvement and contributions of PLWH, adhering to the principles of Greater Involvement of People Living with HIV (GIPA), in the global effort to end the AIDS epidemic [9]. An increasing number of domestic and international researchers have also begun to try to explore different forms of peer support practices in the field of HIV. Cabral designed a set of eight HIV peer education sessions, including knowledge of HIV-related diseases, knowledge of disease self-management, and coping with discrimination [10]. Peer support has been proven effective in enhancing various outcome indicators for PLWH, including mental health, social support, medication adherence, adherence to medical follow-up, and treatment success [11–13]. However, studies have also indicated that peer support often lacks sustainability, resulting in PLWH no longer benefiting after the program concludes [14,15]. Therefore, as a crucial supplement to healthcare services, it is particularly urgent and vital to mobilize further and harness the significant role of peer support in the health management of PLWH.

Our research team has explored the mechanisms of the effectiveness and sustainability of HIV peer support through a realist synthesis [8]. The results indicated that positive communication and atmosphere between peer volunteers and PLWH are essential to ensuring the effectiveness and sustainability of the intervention. A study suggested a psycho social benefit as PLWH connected with peer volunteers, expressed themselves openly [16]. The precise matching of PLWH and peer volunteers can promote a more positive relationship between them so that PLWH are more accepting of peer volunteers' support and that peer volunteers have fewer negative feelings during the peer support process, thus helping to improve the effectiveness and sustainability of HIV peer support [17]. Therefore, the research team conducted research and practice in peer support, grounded in the principle of accurate matching between PLWH and volunteers.

To overcome time and geographical limitations, fully harness and activate the role of peer support in PLWH health management, and encourage more PLWH to participate in peer support, our research team devised a mobile platform to increase the project's reach. The health education and self-management strategies offered through mobile health software have benefited PLWH and healthcare professionals [18]. Nursing professionals play a pivotal role in the holistic care of PLWH, addressing their clinical needs and providing psycho social support, health education, and continuity of care. However, the scarcity of nursing resources and the complexity of PLWH's long-term health management needs often limit the individualized support in traditional healthcare settings. In this context, digital health platforms offer a promising avenue to extend nursing care beyond the clinic, enabling remote monitoring, timely intervention, and sustained patient engagement [19]. Mobile healthcare has wide-ranging applications in the HIV field, and there are already mobile health apps designed for PLWH that provide medication reminders, medical information retrieval, and symptom management [20,21]. To enhance user convenience, rapidly expand the platform's reach to more potential users, and bolster user engagement, the research team promoted the HIV peer support platform as a WeChat Service. As of the end of March 2025, the Chinese multi-functional social media platform had over a billion monthly active users [22], and getting information by following public numbers is a standard function used by WeChat users [23]. As the platform has social attributes, the research team chose the service number rather than the subscription number to facilitate a one-to-one correspondence between platform users and WeChat. In addition, the service number is on WeChat's first-level list, making it more convenient for users to view information from the platform promptly. WeChat users can access and use all of Aspark's features for free by clicking on the "Peer Support" portal after following Aspark's WeChat public number. To our knowledge, few mobile health platforms for PLWH have yet incorporated precision-matching functions. Therefore, we designed "Aspark", a mobile platform that incorporates precise matching for peer support, specifically tailored to provide a mobile hub for HIV peer support and health management for PLWH.

2. Method

2.1. Study design

This study involved two sequential phases: the development of the "Aspark" mobile platform and an evaluation of its usability. A multidisciplinary team designed and built the platform based on a theoretical framework of HIV-related social support, incorporating features such as precision matching, health management tools, and peer support modules. Subsequently, a cross-sectional study using convenience sampling was conducted between March and May 2024 to assess the platform's usability. This study was reported according to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement [24].

2.2. Ethical consideration

This study was conducted according to the World Medical Association Declaration of Helsinki and approved by the Peking University Biomedical Ethics Committee (IRB00001052-23116).

2.3. Development of the platform

2.3.1. Multidisciplinary team and technical implementation

A multidisciplinary research and development team was assembled, including a researcher (project PI), a software engineer, an art designer, two clinical nurse specialists, two full-time HIV peer volunteers who are also PLWH and can provide advice from both the PLWH and volunteer perspectives, and a counselor. Adopting an agile software development approach, the R&D team holds weekly meetings during the software development period to follow up on the development progress and discuss the goals and problems encountered. Software development using the Java/JavaScript programming language, deploying Tencent Cloud servers to collect all the multi-modal data users leave on the platform, including numbers, text, voice, images, videos, etc. Back-end development is based on Vue 3 and the Vite framework, and a unique account password is used to log into the back-end. The People's Republic of China authorized the platform for computer software copyright (2024SR0117803).

2.3.2. Theoretical framework

The software design is based on the foundation of the group's previous theoretical research, and the platform provides five types of social support for PLWH: informational, instrumental, emotional, affiliational, and appraisal support [8]. Information support helps PLWH to understand medical information better. Instrumental support can be divided into medical instrumental and instrumental support for daily living. Emotional support is also an essential component of peer support. Affiliation support aims to help PLWH reduce isolation and connect to social networks, and assessment support aims to help PLWH make decisions by providing information, weighing the pros and cons, and conducting motivational interviews. Guided by confirming cases sampling, we conducted a descriptive qualitative study at Beijing You'an Hospital, conducting in-depth interviews with two clinical nurses, two peer volunteers, and four PLWH. Their narratives allowed us to nuance and enrich every dimension of social support [25]. The insights inform the iterative design of more platform modules and provide a theoretical basis for the relevant video materials within the platform.

Regarding the precise matching mechanism, based on the the social determinants of health framework [26] and interpersonal circumflex model [27], 14 matching variables (interpersonal style, route of infection, presence of commodities or complications, hobbies and interests, $CD4^+$ T-cell counts, residential status, occupation type, age, education level, gender, marital status, duration of antiretroviral medication, duration of HIV infection, and monthly income) were identified for the matching system [28]. The weights of the variables were calculated by combining the precedence chart method [29] and the independence weight method [30]. The weighted eigenvector correlation algorithm was used for similarity calculation. The similarity of interpersonal style variables was converted to "8 to 2" scores based on the theory of interpersonal loop model, i.e., the scores of 8 dimensions were converted to dominance and agreeableness scores, in which the dominance scores were inverted, and then finally combined with the weights of the variables to match the peer volunteers with the highest scores and the best recommended matches [28].

2.3.3. Video production

We established an interdisciplinary video production team comprising a professional videographer, a clinical nurse specialist, an HIV peer volunteer, and a psychological counselor. Grounded in our prior empirical findings [8,25], the team drafted the scripts according to the principles of plain language, conciseness, and case-based illustration; all drafts underwent iterative internal review and refinement. Filming was conducted in the studio and conference room of Beijing Youan Hospital, Capital Medical University, yielding six educational videos. Specifically, the HIV peer volunteer presented segments on social support overview, platform navigation, decision-making, and interactive support; the clinical nurse specialist delivered informational support content; and the psychological counselor addressed psychological and emotional support and decision-making support. The complete video series has been registered with the National Copyright Administration of China (Registration No. 2023-I-00244474).

2.4. Evaluation of the platform

2.4.1. The setting and participants

All users who registered and used the platform for over two weeks are invited to complete the questionnaire voluntarily. Regarding sample size, Bastien's research indicates that 5 to 15 participants can identify most usability issues [31]. This study is expected to include at least 100 participants.

2.4.2. Data collection

This study distributes questionnaires via email to collect electronic questionnaires. The questionnaire comprises a general information questionnaire and a System Usability Scale (SUS). This study used the cross-culturally adapted Chinese System Usability Scale (C-SUS) to assess platform usability. Developed via forward–backward translation, structured interviews, and cognitive pre-testing, the C-SUS demonstrates good reliability (Cronbach's $\alpha = 0.84$, 95 %*CI* [0.807, 0.870]) [32]. The SUS has 10 entries and is scored on a Likert-5 scale (completely disagree - agree), where odd-numbered entries are positive entries and even-numbered entries are negative entries, with 1 point for completely disagree and 5 points for agree. SUS is scored as follows: total SUS score $= (X + Y) \times 2.5$; X = score for all odd entries - 5; Y = 25 - score for all even entries. With this scoring rule, SUS is out of 100 points, and a score of 68 or more is considered good usability for users [33].

2.4.3. Data analysis

The data were entered into SPSS 26.0 software for statistical analysis, and the data entry process was double-checked to descriptively analyze the general situation of the study participants and the SUS scores. Measurement data were statistically described by calculating frequency and percentage, and count data were described by calculating mean and standard deviation.

3. Results

3.1. Description of the platform

3.1.1. Accurate matching function between PLWH and peer volunteers

To subscribe to our service for free, search "Starfire Original" in WeChat. The homepage offers a feature for precise matching between PLWH and volunteers, along with a flip page for key notifications (Appendix A). The first time the button is clicked, the platform invites the user to complete a questionnaire (Appendix B). The questionnaire includes demographic and clinical characteristics, as well as interpersonal styles. The platform uses this data to match PLWH with volunteers accurately.

After clicking the "I need help" button, the background will perform the accurate matching algorithm and recommend volunteers (Appendix C). Users can select a match from the suggested list, and after matching, they can chat and make phone calls, and

volunteers can provide various kinds of support for users (Appendix D).

At the same time, by clicking the "I can provide the help" button, the platform will jump to the interface of volunteer training (Appendix E). Our research group has recorded five training videos corresponding to the knowledge and skills of the five types of social support. When the user has watched all five videos, he/she can obtain the volunteer certificate (Appendix F) and provide support to other users on the platform. In addition, after receiving the certificate, the training videos can be watched repeatedly, and users can record learning notes on the platform (Appendix G). After completing the training, the volunteers have dual identities: PLWH and volunteers. The interaction between PLWH and their peers is not merely a one-way assistance relationship, but a mutual support system where both parties gain from giving and receiving help. This dynamic allows them to acquire more profound knowledge about the self-management of their condition.

3.1.2. Discover page

The Discover page on the "Aspark" platform offers comprehensive tools and features to help users manage their health and well-being. The page includes 10 interactive buttons to provide users with support and management tools such as medication interaction queries, health management tools, medication reminders, and mood diary (Appendix H).

3.1.2.1. Medication interaction queries. This module offers a query function for interactions between 25 ART drugs and 383 other medications (Appendix I). The information on these drug interactions is sourced from pertinent guidelines [34].

3.1.2.2. Health management tools. Users can input their current laboratory test results, and the system offers four pre-set templates for easy data recording. The system automatically generates a trend graph once the user has entered their data. It highlights whether the values fall within the normal range, aiding users in objectively tracking and comprehending their health indicators (Appendix J). Users can add relevant monitoring parameters based on their doctor's advice and personal needs for indicators not included in the templates.

3.1.2.3. Medication reminders. Users can utilize daily medication reminders and a punch-card feature (Appendix K). The button, adorned with an alarm clock icon, assists users in timing their medication intake and evaluating their adherence to the prescribed regimen. It also logs the reasons for missed or incorrect doses, further aiding in the user's medication adherence management.

3.1.2.4. Mood diary. Users can click on a specific date and select "Record the Mood" at the bottom. They can engage in expressive writing by choosing the appropriate emoticon to represent their mood and optionally adding text in the provided text box (Appendix L). This provides a holistic view of the user's emotional state and fluctuations, aiding them in tracking and understanding their mood over time. For example, a user might select a "sad" emoticon and write: "Felt very tired and lonely today after the hospital visit. Worried about the latest test results, but glad I at least took my medication on time." Over time, these entries generate a visual mood timeline, providing a holistic view of the user's emotional state and fluctuations, aiding them in tracking and understanding their mood over time.

3.1.2.5. Post center and forward post center. The "post center" (Appendix M) and "forward post center" (Appendix N) serve as valuable sources of health education information for users. From Monday to Friday, new content is released daily in these sections.

3.1.2.6. Medical housekeeper. In the "medical housekeeper" module (Appendix O) on the Discovery page, users can add their own consultation notes and record their healthcare burden, including direct, indirect, and intangible costs (Appendix O). This feature displays the total amount for each cost type and visually represents the percentage distribution using a fan chart, making it straightforward to comprehend their consultation expenses (Appendix O). Furthermore, users can specify the date and time for their consultation reminder, and the system will alert them to attend the consultation at the designated time.

3.1.3. Community page

Three modules–topic discussion, personal affairs appointment, and activity center–can provide information and emotional and instrumental support for users (Appendix P).

3.1.3.1. Topic discussion. Users on the platform have the flexibility to explore and comment on topics initiated by others, or they can create their own topics (Appendix Q). This feature offers PLWH comprehensive support through information, psychology, emotions, and decision-making assistance. For example, a user can post a topic in the discussion section to ask about the differences between "national free drugs" and "self-paid drugs". Other users can then respond to the post, sharing their experiences, insights, and opinions, providing support and information to the community.

3.1.3.2. Personal affairs appointment. The platform enables users to facilitate assistance among PLWH by publishing and browsing affairs (Appendix R). This platform feature specifically intends to offer instrumental support to medical and non-medical PLWH.

3.1.3.3. Activity center. The module allows users to initiate, view, and participate in offline activities, thereby promoting social interactions among them (Appendix S). This functionality is designed to provide PLWH with social interactive support, aiding their smooth reintegration into society.

3.1.4. Privacy protection mechanism

To avoid users registering duplicate platform accounts, "Aspark" requires WeChat authorization for all social functions (matching, chatting, posting, and commenting) to ensure that one WeChat user corresponds to one "Aspark" user. Authorizing WeChat only provides temporary access to the WeChat avatar and nickname, which cannot locate and decode other sensitive personal information of the user. Users can change their avatars and nicknames on "Aspark" to make their WeChat accounts ambiguous. In addition, all chats on "Aspark" were retained for only seven days to maximize user privacy.

3.1.5. Nursing support and clinical integration

The "Aspark" platform facilitates meaningful nursing involvement in HIV care beyond conventional clinical settings. Nursing staff play a vital role in training and supervising peer volunteers, ensuring that all support interactions are safe, appropriate, and effective. Through the health management module, nurses can remotely monitor trends in patients' laboratory results, such as $CD4^+$ counts and viral load, and track medication adherence. This enables timely feedback and early intervention when abnormalities are detected. The mood diary and community discussion modules offer nurses insight into users' emotional states, allowing them to provide professional psycho social support, motivational

communication, and crisis intervention when needed. Moreover, the structured data collected by the platform—including symptom patterns, medication logs, and activity records—provides valuable information that nurses can use to develop personalized care plans and offer tailored guidance during clinical consultations. This continuous, data-informed engagement allows nursing professionals to deliver holistic, individualized support that complements traditional face-to-face care.

3.2. The step of evaluating the platform

During the data collection period, 130 people submitted online questionnaires on the platform, accounting for 29.5 % of the total registrants in the backend. Among the 130 participants, 97.7 % were male, with a mean age of 31.8 ± 6.6 years, and 46.2 % held a postgraduate degree. Most were single (92.3 %) and employed (90.0 %). Mean time since diagnosis was 5.5 ± 4.5 months; 85.4 % acquired HIV through sexual contact, and the mean CD4 count was 509 ± 144 cells/μL. The number of the 130 people with SUS of 68 or above was 108, which accounted for 83.08 % of the total registrants. Participants consider the usability of this platform is good.

4. Discussion

To our knowledge, “Aspark” is one of the pioneering mobile health platforms designed to provide peer support for PLWH and facilitate a match between PLWH and peer volunteers, offering practical tools for health management. The rapid development of mobile healthcare, which combines the characteristics of traditional medical and healthcare services and the Internet with convenience, time and space irrelevance, and resource sharing, has provided a practical and feasible new method of disease management for a wide range of chronic diseases, including hypertension [35], diabetes [36], and cancer [37]. Meanwhile, by integrating nursing expertise into the design and implementation of mobile health interventions, such platforms can enhance the quality and accessibility of care for PLWH, particularly in resource-limited environments. Thus, the involvement of nursing professionals in developing and operationalizing a mobile health platform like “Aspark” is essential to its effectiveness and sustainability. Mobile healthcare is also becoming more widely used in the HIV field [38]. A meta-synthesis of qualitative research showed that peer support offers a mutually beneficial option for addressing resource allocation and social support for PLWH, and that effective models must be developed for HIV peer support practices [39]. A peer-led mobile app for treatment in HIV (PATH) was used to improve viral suppression among Hispanic and Black PLWH [40]. A study adapted a mHealth intervention to increase acceptance of HIV care among young PLWH [41]. Han et al. designed and developed a Personalized Symptom Management Mobile Health Application for Persons Living with HIV in China, which was a promising and flexible tool for HIV symptom management [42]. It provided PLWH with personalized symptom management strategies and facilitated the case management for medical staff. The dual characteristics of AIDS, both infectious and chronic, determine the long-term and complex nature of the health management needs of PLWH. Therefore, mobile health programs are promising options for addressing PLWH health management issues.

Accurate matching is widely used in medical and other fields, such as recommending suitable self-guided content for patients by matching their basic information and health problems [43] and filtering out resources to meet their needs from many online medical resources [44]. There is also a wide range of applications in non-medical domains, such as interpersonal matching recommendations in social media based on user interests and preferences [45]. To our knowledge, this is the first application of precision matching in HIV peer support. Previous mobile health programs providing social support for PLWH did not consider the matching between volunteers and PLWH [16,36,46,47]. “Aspark” relies on a peer support matching mechanism based on symbolic interaction theory to achieve an accurate match between PLWH and peer volunteers, thus improving the effectiveness and sustainability of support. In addition, “Aspark” relies on WeChat for greater accessibility, more exhaustive coverage, and greater promotion potential. It is embedded in the HIV peer-support training system by integrating peer support and precise matching. Integrating the HIV peer support precise matching mechanism and realizing the accurate matching between PLWH and peer volunteers with the support of symbolic interaction and other theories. “Aspark” constructed a mobile platform and educational platform for HIV peer support to optimize HIV peer support services, fully mobilize the role of HIV peer support in complementing HIV medical resource services, enhance comprehensive individualized health management for PLWH, and enhance the effectiveness and sustainability of peer support.

5. Limitation

Several limitations of this study should be noted. The platform has not yet been connected to the hospital information system, making it impossible to obtain relevant disease information directly from the system. It also affects the promotion and sustainability of the platform to a certain extent. Our research team will work on connecting the platform to hospital information systems and publicizing the platform through social media and academic conferences to expand the user base further. The usability evaluation results indicate that users rate the platform’s usability favorably. However, the study suffers from a small sample size. To ensure that “Aspark” meets the needs of PLWH, peer volunteers, and medical staff, usability evaluations should be scaled up to collect feedback from these stakeholders. Based on this feedback, “Aspark” can be improved and updated to optimize the precision matching mechanism and enhance HIV peer support and health management. Furthermore, rigorous clinical trials should be designed to assess the efficacy of the application of this tool.

6. Conclusion

In conclusion, the “Aspark” mobile platform represents a novel and feasible digital intervention designed to enhance peer support and health management for PLWH through a precision-matching mechanism. By integrating evidence-based social support frameworks, the platform provides comprehensive support—including informational, emotional, instrumental, affiliational, and appraisal support—tailored to the unique needs of PLWH. The high usability ratings from initial evaluations underscore its potential acceptability and practical value.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Yangfeng Wu: Conceptualization, Methodology, Investigation, Writing - original draft, Visualization. **Shuyu Han:** Conceptualization, Methodology, Investigation, Writing - original draft, Visualization, Project administration. **Sikai Shan:** Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Xin Xie:** Conceptualization, Writing - original draft, Writing - review & editing, Visualization. **Dongmei Li:** Conceptualization, Writing - original draft, Writing - review & editing, Visualization. **Lili Zhang:** Conceptualization, Writing - original draft, Writing - review & editing, Visualization. **Zhiwen Wang:** Conceptualization, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT for editorial support. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of competing interest

The authors have declared no conflict of interest.

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Appendices. Supplementary data

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